Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11	(a)	(i)		**large** increase after removing the 2nd electron			1	1		
		(ii)		student is partially correct / incorrect (can be inferred from answer) **–** must have some sensible attempt at explanation (1) **X** is higher than **Y** since **X** has a larger nuclear charge but little/no extra shielding (1) **X** is not lower than **Z** since the outer electron in **Z** is partly shielded by the s electrons (and therefore more easily lost) (1) reference to stronger attraction is neutral			3	3		
	(b)	(i)		$Na^+(g) \rightarrow Na^{2+}(g) + e^-$	1			1		
		(ii)		$\Delta E = \frac{4560000}{6.02 \times 10^{23}} = 7.57 \times 10^{-18}$ J (1) $f = \frac{\Delta E}{h} = \frac{7.57\times10^{-18}}{6.63\times10^{-34}} = 1.14 \times 10^{16}$ s^{-1} (1) $\lambda = \frac{3.00 \times 10^{8}}{1.14 \times 10^{16}} = 2.63 \times 10^{-8}$ m (1) λ = 26.3 nm (1) ecf possible throughout	1	3		4	4	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		award (1) for each of following H—C—H 109.5° accept 109° Cl—Al—Cl 120° Cl—Be—Cl 180°	3			3		
		(ii)		(second sentence is incorrect because) total number of electron pairs important not just bonding pairs (1) e.g. bond angle in H_2O (105°) is less than that in $AlCl_3$ (120°) even though there are fewer bonds in H_2O (1) accept any sensible example but must refer to two molecules / structures / shapes / angles for second mark			2	2		
				Question 11 total	**5**	**3**	**6**	**14**	**4**	**0**